Muhammad Ariff Bin Abdul Razak

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Systems Engineer with a foundation in robotics, embedded systems, and machine learning. Twenty-two months in the Singapore Police Force's Protective Security Command gave a first-hand view of how Home Team operations run, and the work since has stayed close to those problems — benchmarking AI-generated media detectors as a public-safety threat, and now leading a quiet-drone capstone with MHA as its end user. Drawn to engineering where reliability is the point, and to contributing that to public safety directly.

WORK EXPERIENCE

KLASS Engineering and Solutions
Machine Learning & Fullstack Engineer (Intern)
Sep 2025 – Aug 2026

- Reproduced and benchmarked state-of-the-art AI-generated video and audio detectors (NSG-VD, D3, ReStraV, SpeechForensics) on SLURM-based HPC to assess detection capability against AI-generated media as a public-safety threat, achieving 90.03 AUROC on the NSG-VD replication.
- Ran ablation studies integrating optical flow (RAFT), segmentation (SAM 2), and point tracking (CoTracker) to test whether physical-motion signals in video sequences can distinguish AI-generated content, finding mixed evidence of a usable signal.
- Engineered a data generation pipeline chaining video extension (Wan2.2), voice cloning (Chatterbox), and lip-sync (LatentSync) models to curate a balanced 1,200-video real/fake dataset, used to compare all four detectors consistently and to test models at small scale before committing larger compute.
- Managed GPU-accelerated experimentation on shared SLURM clusters (NVIDIA H200 nodes) via Docker containers, running sanity-check protocols on custom validation splits that caught pipeline errors before committing to training and synthetic-deepfake generation runs of up to 5 hours each.
- Built React frontend components and FastAPI backend services with WebSockets for real-time streaming responses, deploying multi-model AI endpoints for a company-wide internal AI chatbot platform that gave staff a secure alternative to public AI tools, with search across internal documents.

Singapore Institute of Technology (SIT)
Student Research Engineer
Aug 2024 – May 2025

- Designed and built a temperature-controlled enclosure around a compression testing machine for faculty research on how foods behave during chewing, defining the electrical architecture and selecting components before handing the completed chamber over to the research team.
- Engineered closed-loop temperature control with a thermistor and standalone temperature controller, holding the chamber within ±2 °C of a 37 °C body-temperature setpoint for consistent test conditions.

SUTD – ROAR Lab
Research Assistant
Aug 2022 – May 2024 (Volunteer until Nov 2022, Full Time thereafter)

- Led the design of Panthera's intelligent lighting control system to signal the autonomous vehicle's intent to pedestrians and meet testing safety requirements, running a rosserial node on an Arduino under an RTOS that mapped cmd_vel, IMU, and vehicle-state topics to turn, warning, and hazard signals; the platform passed T1 autonomous testing at CETRAN.
- Wired and integrated the 3D LiDAR used for Panthera's obstacle detection, mapping, and localization, verifying sensor output through depth-map visualization, and supported the vehicle through its T1 autonomous field testing.

Singapore Police Force – Protective Security Command
Trooper & Armourer
Jan 2021 – Nov 2022 (Full-Time National Service)

- Performed protective security duties within a deployed team at strategic locations and major national events, operating under strict standard operating procedures across 22 months of full-time service.
- Served as unit Armourer alongside trooper duties, managing the issuance, return, and accountability of arms and equipment with zero discrepancies, and maintaining audit-ready records under regulated inventory procedures.

SOLUSTAR
Robotics Engineer Intern (Unpaid, Part Time)
May 2022 – Aug 2022

- Designed and debugged PCBs and circuitry for a disinfectant robot, bringing the boards into a working prototype.

EDUCATION

DigiPen Institute of Technology & SIT
Aug 2023 – May 2027

- B.Eng. Mechatronics Systems
- Relevant Coursework: Mechanics, Control Systems, Robotics, Embedded Systems, Machine Learning, Systems and Software Engineering

Ngee Ann Polytechnic
Sep 2022 – Sep 2023

- Specialist Diploma in Robotics Engineering (SGInnovate PowerX)
- Relevant Coursework: Autonomous Robot Navigation, Machine Learning/Deep Learning for Robotics, Computer Vision

Singapore Polytechnic
Apr 2017 – Sep 2020

- Diploma in Mechatronics and Robotics
- Relevant Coursework: Microcontroller Applications, PLCs, and Robotics Integration

PROJECTS

1. Capstone (MET4004) — Acoustic Noise Reduction for Drones
Project Lead, Software & Electronics (Sponsored by KLASS)
Aug 2026 – Apr 2027

- Leading a 3-person team to deliver a proof-of-concept quiet indoor drone for end user MHA to reduce acoustic detectability, targeting a 78 dBA ceiling at a 1 m reference during steady hover, 10-minute hover endurance, and a 370 mm square airframe envelope, with the project proposal approved.
- Owning the software and electronics workstream, selecting a low-noise propulsion stack of 5-inch injection-molded toroidal propellers, custom ducts with acoustic lining and serrated trailing edges, 1300 KV low-RPM motors, and AM32 ESC PWM frequency tuning to shift tonal whine into less perceptible bands.
- Specified the avionics and power architecture (Holybro Pixhawk 6C Mini, PM02 V2 power module, dedicated 150 A power distribution board), yielding a derated 3.3:1 thrust-to-weight ratio at 1,200 g all-up weight to preserve control headroom for manually piloted indoor flight.
- Designed the acoustic validation plan to the Von Karman Institute four-microphone methodology with TU Berlin distance normalization to a 1 m reference, to benchmark directly against HKUST's 90.52 dBA commercial quadcopter baseline and VKI's DJI Phantom 3 SE measurements.
- Specified psychoacoustic post-processing in Python (MoSQITo) under both Zwicker (ISO 532-1) and Moore–Glasberg (ISO 532-2) loudness standards to quantify perceived annoyance beyond raw sound-pressure level.

2. Autonomous Airport Trolley Return Robot (SIT–DigiPen Coursework)
Firmware & Perception

- Built LiDAR-based perception in C++ and ROS for a robot that autonomously returns airport trolleys, developing and integrating obstacle detection, SLAM mapping, and localization with IMU and wheel-odometry sensor fusion on an onboard computer, completing the final demo task.
- Developed RTOS firmware on an STM32F7 microcontroller with concurrent tasks for obstacle detection and joystick control, and a custom serial protocol linking it to ROS for reliable motor control.

SKILLS / COMPETENCIES

AI, Vision & Audio Analytics: Deep Learning, Computer Vision, Multimodal ML, Deepfake Detection, Generative AI, OpenCV, Optical Flow (RAFT), Segmentation (SAM 2), Point-Tracking (CoTracker)
Machine Learning Frameworks: PyTorch, TensorFlow, Keras, Hugging Face Transformers, CNNs, DNNs, LSTMs, YOLO, Vision Transformers, Reinforcement Learning, Model Benchmarking, Zero-shot Learning
Robotics, UAV & Autonomous Systems: Embodied AI, ROS, SLAM (3D LiDAR), Perception, Sensor Fusion, Visual-Inertial Odometry (VIO), Autonomous Mobile Robots, Quadcopter Design, UAV Flight Testing, UAV Aeroacoustics, Acoustic Noise Reduction, PX4 Autopilot, ArduPilot, Pixhawk
Embedded, Hardware & Test: Embedded C / C++, RTOS, Microcontrollers (STM32, Arduino), PID Control, Closed-Loop IoT Control, PCB Design, Firmware Development, Hardware Prototyping, Experimental Design, Test & Validation
Software & Fullstack: Python (NumPy, Pandas), C++17, C#, Java, TypeScript, React, Zustand, FastAPI, OpenAPI, WebSockets, Git, Linux/Ubuntu, WSL

AWARDS / ACHIEVEMENTS / CERTIFICATES

- MOE (Ministry of Education) Instructor Certification (Achieved at age 18) — Valid Oct 2018 – Oct 2026 (renewing)

VOLUNTEERING & COMMUNITY SERVICE

ROSCon Singapore 2025 (Open Robotics) — Volunteer
Oct 2025

- Supported event operations, logistics, and attendee experience at ROSCon Singapore 2025, the annual ROS developer conference with ~1,000 participants from 52 countries, including helping run 3 workshops.